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20 February 1967
Interim Revision of
Fed. Spec. U-R-179
15 September 1954

INTERIM FEDERAL SPECIFICATION

RESIN, ACRYLIC, DENTAL

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal Specification as a valid exception to Federal Specification U-R-179.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers acrylic base denture resin (polymer) suitable for use in dental prosthetic procedures.

1.2 Classification. Denture base resins covered by this specification shall be of one class (powder and liquid), and of the following types and colors, as specified (see 5.1 and 6.2).

Type I. Heat-curing polymers.
Color A. Pink.
Color B. Clear.

Type II. Cold-curing repair polymers.
Color A. Pink.

FSC 6520

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2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

STANDARDS

Federal

FED-STD-123	Marking for Domestic Shipment (Civilian Agencies).

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D.C.

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(Copies of Military specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2. Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN DENTAL ASSOCIATION

A.D.A. Specification No. 12 for Denture Base Polymer.

A.D.A. Specification No. 13 for Denture Cold-Curing Repair Resins.

(Application for copies should be addressed to the Council of Dental Research, American Dental Association, 211 East Chicago Avenue, Chicago, Illinois 60611.)

3. REQUIREMENTS

3.1 Materials. Resin (polymer) shall be supplied in a package containing a bottle of powder and a bottle of liquid.

3.1.1 Liquid. The liquid shall be a methyl methacrylate monomer with a suitable storage polymerization inhibitor. The liquid shall be clear, colorless, and substantially free from undissolved or particulate matter when examined, without accessory magnification, against a black background and against a white background. Illumination shall be from a light which, at a point 10 inches from its source, provides an intensity of illumination of not less than 100 and not more than 350 foot candles. The liquid shall be a cross-link type monomer, which produces a cross-linked cured resin on polymerization with the powder, when processed in accordance with manufacturer's instructions.

3.1.1.1 Stability. The liquid shall show no thickening, when tested as specified in 4.4.1.1.

3.1.1.2 Discoloration. There shall be no visible discoloration of the exposed sample, when tested as specified in 4.4.1.2.

3.1.2 Powder. The powder shall be poly (acrylic acid esters), poly (substituted acrylic acid esters), or copolymers or mixtures of the foregoing. The powder shall be free of extraneous material such as dirt or lint.

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3.2 Properties of uncured polymer. The uncured polymer, when used as directed by the instructions accompanying the package (see 3.4), shall be considered ready for packing when the plasticity is such that the polymer separates cleanly from the walls of the glass mixing jar (see 4.4.2.1). This condition shall be attained in 15 minutes or less from the start of mixing for Type I polymer, and 20 minutes or less for Type II polymer. When subjected to the die intrusion test specified in 4.4.2.2, the mixed polymer shall be capable of being intruded into at least two holes of the die (fig. 1) to a depth of not less than 0.5 millimeter (mm.).

3.3 Properties of cured polymer. Type I polymer, when processed according to instructions furnished by the manufacturer, shall yield a satisfactory denture. Type II polymer shall yield a satisfactory denture repair. Properties of the cured polymer shall be as specified in 3.3.1 through 3.3.10.

3.3.1 Glossy surface and defects. The polymer shall have a glossy surface and shall show no surface defects, when cured in a gypsum mold lined with tin foil.

3.3.2 Polish. The polymer shall present a smooth surface having a high gloss, when polished by the conventional dental methods.

3.3.3 Toxicity. The manufacturer shall submit a statement that the denture base produced according to the instructions supplied with the material has no toxic effect on the normal, healthy person.

3.3.4 Color. The depth of color of the pink polymer shall be equivalent to the standard sample (see 6.4). The color shall be uniformly distributed. Clear polymer shall be colorless.

3.3.5 Transparency and translucency. The specimen plate of clear polymer, formed as described in 4.4.6, shall be sufficiently transparent to read clearly this paragraph through the plate. The specimen of pink polymer, formed as described in 4.4.6, shall be translucent.

3.3.6. Porosity. The specimen plate (fig. 2) cut as described in 4.4.6 shall show no bubbles or voids.

3.3.7 Sorption. The increase in weight of the polymer shall be no more than 0.7 milligram per square centimeter (mg/sq. cm) when tested as specified in 4.4.4.

3.3.8 Solubility. The loss in weight of the polymer shall be no more than 0.04 mg/sq. cm. of surface, when tested as specified in 4.4.5.

3.3.9 Transverse deflection. The deflection of at least three of the five specimens prepared as outlined in 4.4.6 shall meet the requirements in table I.

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Table I. Transverse Deflection Limits

Polymer	For 1500 gm. to 3500 gm. load:* Maximum deflection, mm.	For 1500 gm. to 5000 gm. load:* Minimum deflection, mm.	Maximum deflection, mm.
Type I	2.0	2.0	5.0

Polymer	For 1500 gm. to 2500 gm load:* Maximum deflection, mm.	For 1500 gm. to 4000 gm. load:* Minimum deflection, mm.	Maximum deflection, mm.
Type II	1.5	1.0	4.5

* Deflections shall be measured from an initial load of 1500 grams (gm.) to the load indicated.

3.3.10 Color stability. The polymer shall show no more than a slight change in color when tested as specified in 4.4.7.

3.4 Manufacturer's instructions. Instructions for storing and processing the polymer shall accompany each package. Two copies of all instructions shall be forwarded to the contracting officer for information purposes.

3.4.1 Type I. The instructions for Type I polymer shall include the powder-liquid ratio, time of mixing, time at which the polymer attains the proper plasticity at 23.0 ± 2.0° Centigrade (C.), temperature of flask when the polymer is packed, time to bring polymer to recommended curing temperature, time to hold at curing temperature, method of cooling flask, and time to release flask. Manufacturer's instructions shall conform to those specified in 4.4.3. In addition to the instructions given in 4.4.3 the manufacturer may provide one alternate set of instructions. The manufacturer shall certify that the polymer will meet the requirements of this specification when it is processed according to the alternate method, as well as according to the prescribed methods.

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3.4.2 Type II. The instructions for Type II polymers shall be those prescribed by the manufacturer. Adequate and accurate instructions for using the polymer as a repair material shall accompany each package. The manufacturer's instructions shall include a method for repairing the polymer.

3.5 Certification. The polymer shall be tested at the National Bureau of Standards for the purpose of establishing a list of certified materials. At time of award of contract, the manufacturer shall furnish, upon request, a certificate and data showing that the brand of material to be furnished is the same in quality and brand as that appearing on the list of certified materials. Compliance with this certification requirement does not absolve the manufacturer from complete compliance with the other requirements of this specification. Type I polymer shall comply with the requirements of American Dental Association (A.D.A.) specification No. 12; type II shall comply with No. 13.

3.6 Stability. The polymer shall comply with the specification requirements for at least two years from date of shipment, when stored at 50° to 80° Fahrenheit (F.). No reference to shelf life or expiration date shall be included in the instructions or other accompanying data.

3.7 Delivery schedule. Not more than two months shall have elapsed from the date of manufacture of the item to the date of delivery to the Government.

3.8 Workmanship. Workmanship shall be first-class throughout. The powder and liquid shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The Acceptable Quality Levels (AQL's) shall be as indicated in table II.

Table II. Acceptable Quality Levels

Type defect	Inspection level	AQL (percent defective)
Major A	II	N/A (Acceptance number zero)
Major B	II	1.0
Major C	S-1	1.0
Minor	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted on each lot or control number in accordance with table III. If any test determination for a lot does not comply with the requirements specified, the lot shall be rejected.

Table III. Sampling criteria.

Characteristic	Requirement	Test procedure	Number of determinations per lot or batch number
Stability of liquid	3.1.1.1	4.4.1.1	2
Discoloration	3.1.1.2	4.4.1.2	2
Packing readiness	3.2	4.4.2.1	2
Die intrusion	3.2	4.4.2.2	2
Glossy surface	3.3.1	*	2
Polish	3.3.2	*	2
Color	3.3.4	*	2
Transparency and translucency	3.3.5	4.4.6	1 **
Porosity	3.3.6	4.4.6	1 **
Sorption	3.3.7	4.4.4	1 ***
Solubility	3.3.8	4.4.5	1 ***
Transverse deflection	3.3.9	4.4.6	1 **
Color stability	3.3.10	4.4.7	1
Net content of powder and liquid	5.1.1	S.M.I.	1
Compatibility of container	5.1.3.1.3	4.4.8	2

* The polymer processed according to instructions furnished by the manufacturer (3.4).
** Each determination shall be the average of 5 specimens.
*** Each determination shall be the average of 2 specimens.

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4.3 Examination. The polymer shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Polymer shall be examined for defects, including but not limited to those listed in table IV.

Table IV. Classification of Defects.

Categories	Defects
Critical:	None defined.
Major A:	
101	Type not as specified.
102	Color not as specified.
103	Polymer material not as specified.
104	Unit not complete with powder and liquid.
105	Powder not packaged in wide-mouth glass bottles of suitable size.
106	Liquid not packaged in narrow-mouth, light-resistant glass bottle of suitable design.
107	Closure not of type specified.
108	Manufacturer's instructions do not include: a. Powder-liquid ratio. b. Temperature of flask when the polymer is packed.(Type I) c. Time to bring to recommended curing temperature.(Type I) d. Time to hold at curing temperature.(Type I) e. Method of cooling flask.(Type I) f. Time to release flask. g. Method for using the polymer as a repair material.(Type II)
109	Labeling not complete: a. Bottles not commercially identified. b. Bottles not marked with lot number. c. The word "FLAMMABLE" not present on unit label. d. Unit carton and intermediate carton not marked as specified.

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Major B:		
151		Glass containers cracked or broken.
152		Liquid is turbid (not water clear).
153		Liquid not free of deposits or sediment (see 3.11).
154		Powder not free of extraneous material such as dirt and lint.
155		Instructions not included.
156		Cap does not have nonabsorbent lining material.
157		Liquid leaks from cap.
158		Inner seal does not adhere to mouth of bottle.
159		Inner seal not present on liquid and powder containers.
Major C:		
176		Bottles in unit carton not secured against shifting or damage or accidental opening of package.
Minor:		
201		Labeling not legible, not free from defacing marks and scratches.
202		Lining material does not remain in cap when cap is removed.
203		Cap is not tight.
204		Unit package not as specified.

4.4 Tests. Equipment and material shall be conditioned at 23.0 ± 2.0° C. for not less than 10 hours prior to the tests being made.

4.4.1 Tests on liquid.

4.4.1.1 Stability. The liquid shall be heated in the dark in a closed container at 60 ± 2° C. for 24 hours. Examination shall be made for compliance with 3.1.1.1.

4.4.1.2 Discoloration. Discoloration of the liquid shall be determined by placing 10 milliliters (ml.) of the liquid into a clear glass test tube, having an inside diameter of approximately 1 cm., and by exposing this sample to bright diffuse sunlight for 24 hours. This sample shall be compared with 10 ml. of the unexposed liquid placed in a similar test tube. The tubes shall be viewed perpendicularly to their long axis when backed by a white background for compliance with 3.1.1.2.

4.4.2 Tests on uncured polymer.

4.4.2.1 Packing readiness test. Polymers shall be mixed as directed by the instructions accompanying the package, except that the mixing temperature shall be 23.0 ± 2.0° C. The polymer shall be considered ready for packing when the plasticity is such that the polymer separates cleanly from the walls of the glass mixing jar.

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4.4.2.2 Die intrusion test. A six to ten gm. sample of polymer shall be placed on the perforated brass die (fig. 1) to a depth of approximately 5 mm. The die and specimen shall be transferred immediately to an air chamber in which all the equipment has been preheated to within ± 5.0° C. of the packing temperature specified by the manufacturer's instructions (3.4). A glass plate approximately 5 mm. thick and 50 mm. square and a 5,000 gm. weight shall be placed on the specimen immediately. A sheet of polyethylene shall be used as separator between the glass plate and the specimen. The packing test for Type I polymers shall be started 10 minutes after the proper plasticity has been reached; for Type II polymers the time shall be 3 minutes. Fifteen minutes later for Type I polymers, or ten minutes later for Type II polymers, the die and the specimen shall be removed from the air chamber and the polymer stripped from the die. The depth of intrusion shall be measured with a steel scale and magnifying glass or with a dial gauge.

4.4.3 Preparation of specimens of cured polymer for sorption, solubility, transverse deflection and color stability tests. Specimens of Type I polymer shall be processed in accordance with the following schedule; Ten minutes after the proper plasticity specified in 3.2 has been reached, the flasked specimen (fig. 2) held in a flask clamp without springs, shall be submerged in water at 73 ± 1° C. for 1-1/2 hours and then immersed in boiling water for 1/2 hour. When the heating schedule has been completed, the flask in the clamp shall be cooled in air at 23 ± 10° C. for 30 minutes and then immersed in water at 23 ± 10° C. for 15 minutes. For Type II polymers the manufacturer's instructions shall be followed.

4.4.4 Sorption. The water sorption test shall be made in duplicate on disks 50 ± 1 mm. in diameter and 0.5 ± 0.1 mm. thick (fig. 3). These specimens shall be made in a stainless steel die (fig. 3). The die shall be mounted with gypsum in the flask in the same relative position as the specimen plate in figure 2. The gypsum in the shallow half of the flask shall be poured against a glass plate to obtain a smooth surface. Trial packing shall be conducted with moist cellophane used as a separator. Before final closure of the flask, a sheet of tinfoil shall be substituted for the cellophane. All surfaces of the disk shall be smooth, and the top and bottom shall be flat. The disks shall be dried in a desiccator containing anhydrous calcium sulfate ($CaSO_4$) at 37° ± 2° C. for 24 hours, removed to a similar desiccator at room temperature for 1 hour, then weighed with a precision of 0.2 mg. This cycle shall be repeated until constant weight is attained, that is, until the weight loss of each disk is not more than 0.5 mg. in any 24-hour period. The disks shall then be immersed in distilled water at 37.0° ± 1.0 C. for 24 hours. Then the disks shall be removed from the water with tweezers, wiped for 30 seconds with a clean, dry hand towel, waved in the air for 15 seconds, and weighed 1 minute after removal from water. The value for water sorption shall be calculated as follows for each disk:

$$\frac{\text{weight after immersion - conditioned weight (mg.)}}{\text{surface area (sq. cm.)}} = \text{sorption (mg. per sq. cm.)}$$

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The average of the determined values for the two disks shall be recorded to the nearest 0.01 mg. per square centimeter. The final value (average of two determinations) shall be rounded to the nearest 0.1 mg. per square centimeter. When the final value falls midway between the two numbers, the even number shall be recorded.

4.4.5 Solubility. After the final weighing (4.4.4), the disks shall be reconditioned to constant weight in the desiccator at 37 ± 2° C., as was done in 4.4.4. The soluble matter lost during immersion shall be determined to the nearest 0.01 mg. per square centimeter, as follows, for each disk:

$$\frac{\text{conditioned weight (mg.)} - \text{reconditioned weight (mg.)}}{\text{surface area (sq. cm.)}} = \text{solubility (mg. per sq. cm.)}$$

The final value (average of two determinations) shall be rounded to the nearest 0.01 mg. per square centimeter. When the final value falls midway between two numbers, the even number shall be recorded.

4.4.6 Transverse deflection tests. A plate of polymer shall be processed in a gypsum mold lined with tinfoil (fig. 2). Specimens for the transverse test shall be machined from this plate to the dimensions shown in figure 4. The plate (fig. 2) shall be sawed lengthwise into five equal strips with a band saw operated at a speed of approximately 100 meters per minute (328 feet per minute). For Type II polymers the plate shall be at least 2 days old prior to sawing. The strips may then be sanded slightly on the molded surfaces, if necessary, to remove imperfections. All subsequent milling operations shall be with a 50 mm. diameter flycutter, operating at a spindle speed of approximately 1,800 revolutions per minute and at a table feed of about 6 inches per minute. Rough cuts shall be approximately 0.08 mm. deep, and finish cuts shall be 0.03 mm. inch deep. The strips shall be mounted side by side in a vise, and the edges machined parallel in order to remove all band saw marks. The strips at this point shall be oversize in width. One or more strips shall be mounted in a vise and the faces milled to the required thickness of 2.50 ± 0.03 mm. All five strips shall then be clamped face to face in a vise and the edges milled to the required width of 10.00 ± 0.03 mm. The specimens shall be stored in distilled water at 37.0° ± 1.0° C. for 2 days immediately before testing. The specimen shall be mounted in an appropriate instrument, such as is shown in figure 4. An initial load, consisting of the free-moving parts of the apparatus (fig. 4) plus sufficient lead shot to make a load of 1,500 gm., shall be placed on the specimen. Each 500 gm. increment of shot shall be added at a uniform rate during the last 30 seconds of each minute. The schedule shown in table V shall be used in applying the loads and in making the observations:

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Table V. Schedule of loading and reading.

Time		Load	Dial reading*
min.	sec.	gm.	mm.
0	00	1,500	
0	30	1,500	A
1	00	2,000	
1	30	2,000	
2	00	2,500	
2	30	2,500	
3	00	3,000	
3	30	3,000	
4	00	3,500	
4	30	3,500	B
5	00	4,000	
5	30	4,000	
6	00	4,500	
6	30	4,500	
7	00	5,000	
7	30	5,000	C

Deflection at 3,500 gm. = B-A
Deflection at 5,000 gm. = C-A
* Readings shall be made to the nearest 0.01 mm.

The differences between the deflections at the initial load and at specified loads shall be taken as the deflections of the specimen. The recorded deflections of the specimen. The recorded deflections shall be reported to the nearest 0.1 mm. When the determined number falls midway between two numbers, the even number shall be recorded.

4.4.7 Color stability. One of the disks used in 4.4.4 and 4.4.5 shall be exposed to the radiation of a lamp (fig. 5) with an S-1 bulb which has been in use not less than 50 nor more than 400 hours. The light source shall be a combination tungsten filament mercury arc enclosed in Corex D or other glass which has a low transmission below 2,800 Angstrom units. The lamp shall be rated at 400 watts. The specimen shall be carried on an aluminum disk, which shall be centered under the S-1 bulb, and which shall rest on a phonograph turntable, operated at 33 revolutions per minute. The specimen shall be held by screws approximately 0.2 in. above the disk and 5 in. from the center, and shall be maintained between 60 and 65° C. The plane of the top surface of the specimen shall be 7 in. from the bottom of the S-1 bulb. After exposure for 24 hours, the specimen, compared with an unexposed duplicate specimen, shall show not more than a slight change in color (perceptible with difficulty). The comparison shall be made by visual inspection in daylight.

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4.4.8 Compatibility of container. The inner seal shall be removed. The container shall be allowed to stand capped for 24 hours, after which it shall be examined for evidence of incompatibility.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue.

5.1.1.1 Type I. One package consisting of one container of powder having a minimum net weight of 210 grams, and one container of liquid containing 10 percent in excess of the quantity required to combine with the powder, constitute one unit (see 1.2 and 6.2).

5.1.1.2 Type II. One package consisting of one container of powder having a minimum net weight of 250 grams, and two containers of liquid containing six fluid ounces each (minimum), constitute one unit of issue. (see 1.2 and 6.2)

5.1.2 Packaging quantities. The number of units of issue indicated in table VI shall be packaged in each unit and exterior container, as applicable for the required level of protection specified in procurement document.

Table VI Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	Not required	24 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Immediate containers. Powders shall be packaged in wide-mouth glass bottles of suitable size and design. Liquids shall be packaged in narrow-mouth, light-resistant glass bottles of suitable size and design.

5.1.3.1.1 Closure for bottles of powder. Bottles containing powder shall be closed by means of a metal continuous-thread screwcap, treated inside and outside to resist corrosion. Cap shall have a tight-fitting, impervious, resilient liner which must remain in cap when cap is removed from bottle. Liner shall be vinylite or aluminum faced. As an alternate, paraffin faced liner is permitted. Cap shall be tightened to prevent leaking. Bottle shall be further sealed by an inner seal which shall cover and adhere to mouth of bottle, capable of preventing loss of contents should the cap become loosened.

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5.1.3.1.2 Closure for bottles of liquid. Bottles containing liquid shall be closed by means of a continuous-thread screwcap, fabricated of 1.25 hot-dipped tinplate or 0.25 electrolytic tinplate. Screwcap shall be treated inside and outside with a clear lacquer to resist corrosion. Cap shall have a tight-fitting, impervious, resilient liner which must remain in cap when cap is removed from bottle. Liner shall be aluminum faced. Cap shall be tightened to prevent leaking. Bottle shall be further sealed by an inner seal, which shall cover and adhere to mouth of bottle, capable of preventing loss of contents should the cap become loosened.

5.1.3.1.3 Compatibility of container. The container, closure, and liner shall not interact physically or chemically with the substance contained so as to be altered or to alter the strength, quality, or purity of the container contents.

5.1.3.2 Unit package. Each unit of issue shall be packaged in a carton constructed in accordance with PPP-B-636, type CF, class domestic or PPP-B-676. Commercial colors will be acceptable when paperboard cartons are utilized. Bottles shall be protected by flexible corrugated fiberboard sleeves or by suitable interior fittings in such a manner as to secure the bottles in a vertical position and to insure against shifting or damage. Each carton shall be adequately secured to prevent accidental opening.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4, of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

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5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic and method I closure shall be utilized.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3. Marking.

5.3.1 Civil agencies. Shipments shall be marked in accordance with FED-STD-123.

5.3.2 Military agencies.

5.3.2.1 Interior packages. Each bottle shall be commercially identified and shall show the lot (control) number. Each bottle of liquid shall also be marked with the word "Flammable". Each unit carton shall be marked with stock number, item identification, type (i.e., powder and liquid), lot (control) number, type of resin (e.g., if the powder is a mixture of methyl and ethyl methacrylate it shall be stated), date of manufacture (month, year), unit, quantity, name of contractor, and the legend: "Store in a cool, dark place."

5.3.2.2. Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) number and date of manufacture shall be shown. Date packed shall not be required. Marking shall include the legend:

"STORE IN A COOL PLACE (50° - 80° F.)."

6. NOTES

6.1 Intended use. Type I polymer shall be suitable for preparing plastic denture bases. Type II, polymer shall be suitable for repairing plastic denture bases.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Type and color required (see 1.2 and 5.1).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

6.4 Standard sample. For access to standard samples address the procuring activity (see 3.3.4.).

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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BRASS DIE FOR PACKING TEST

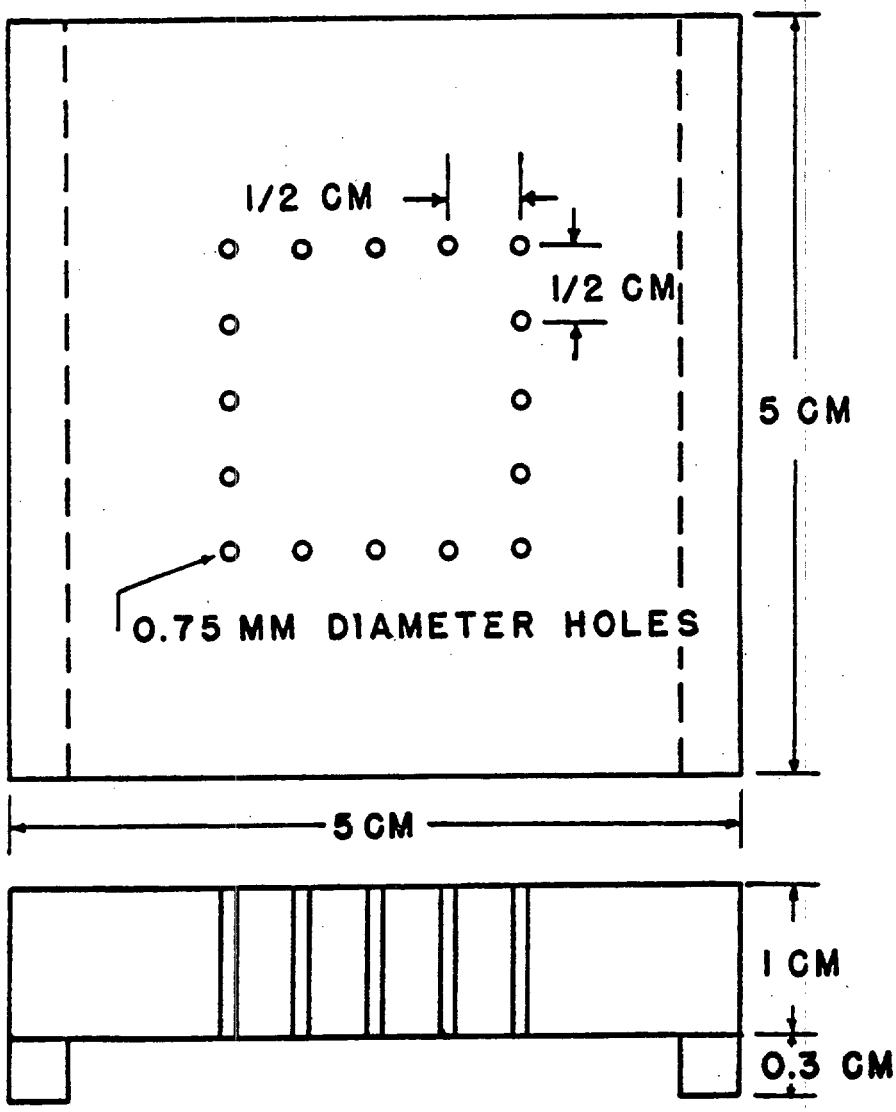
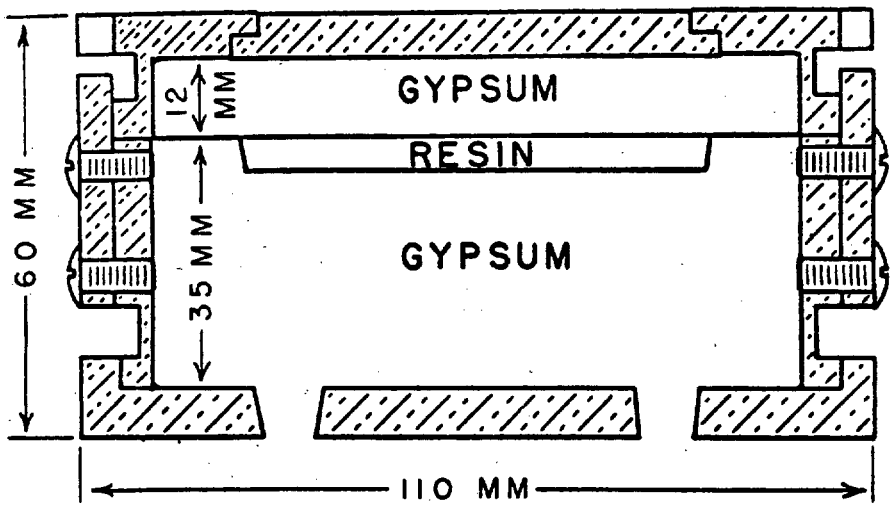
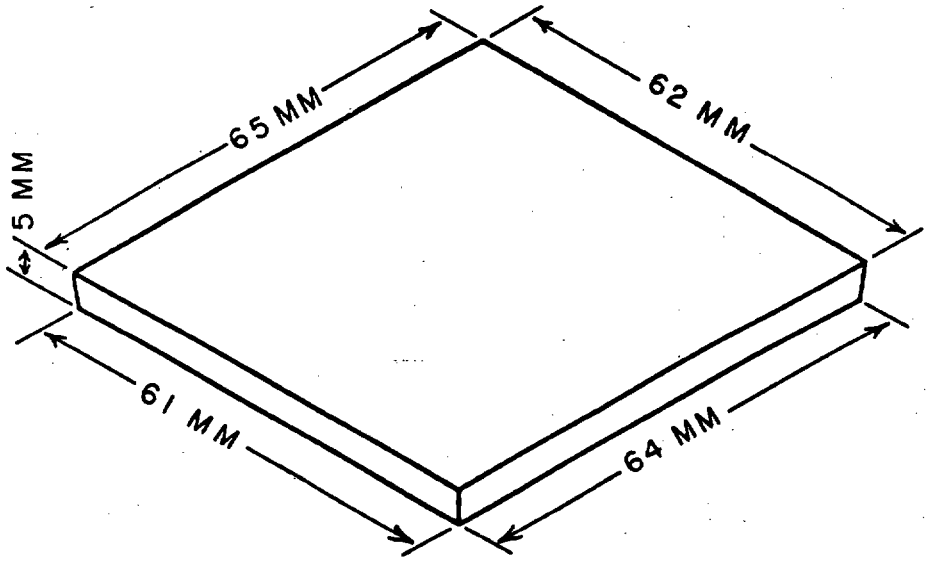


Figure 1

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DENTURE FLASK



SPECIMEN PLATE

Figure 2

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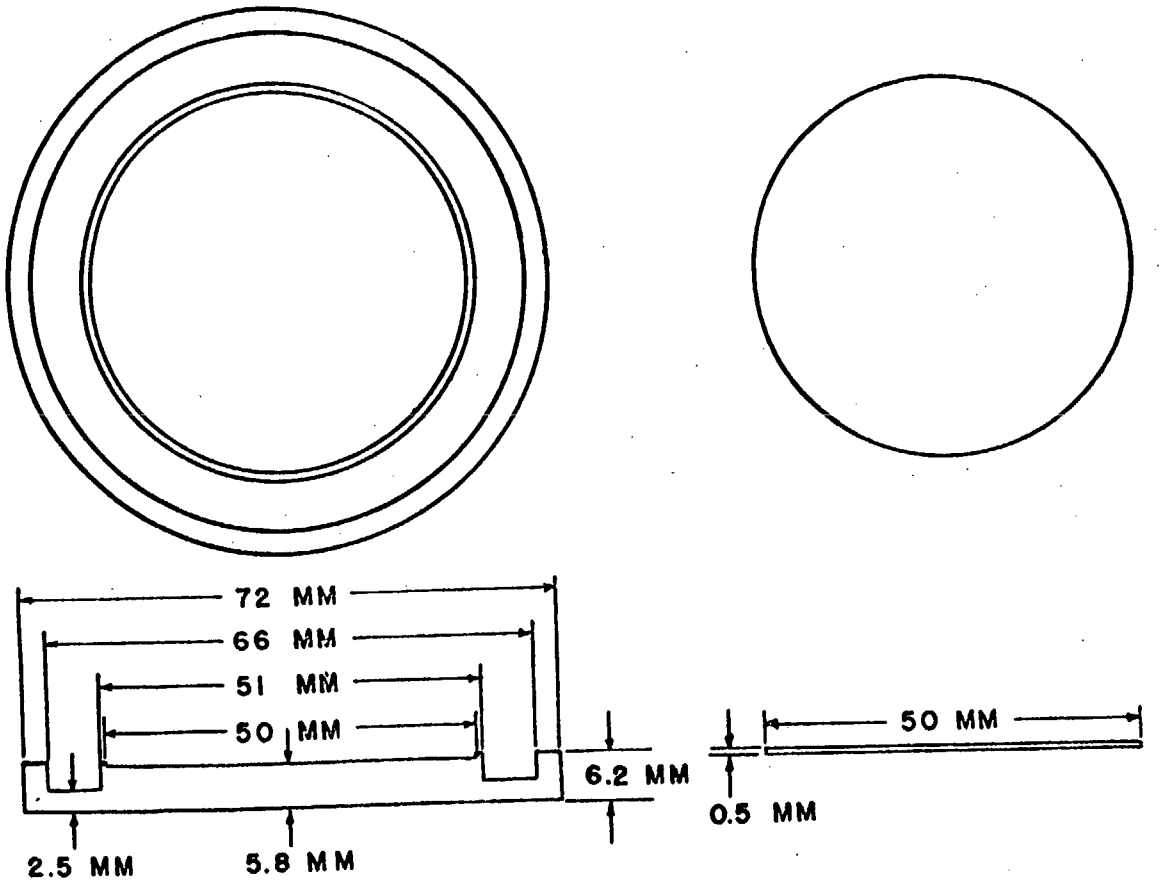


Figure 3

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TRANSVERSE TESTING MACHINE

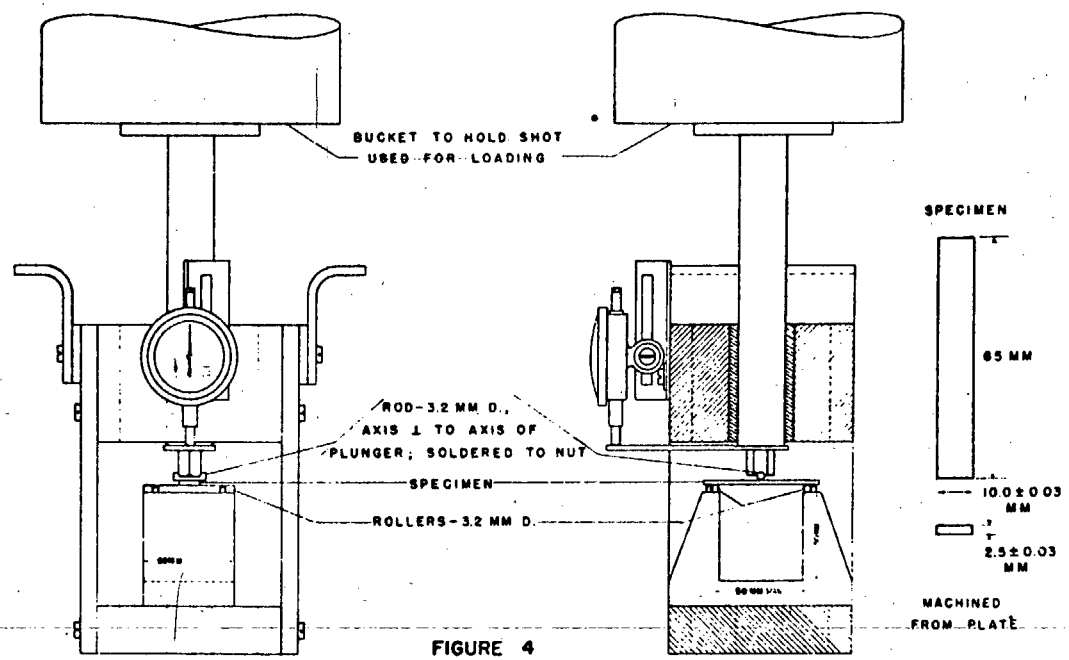
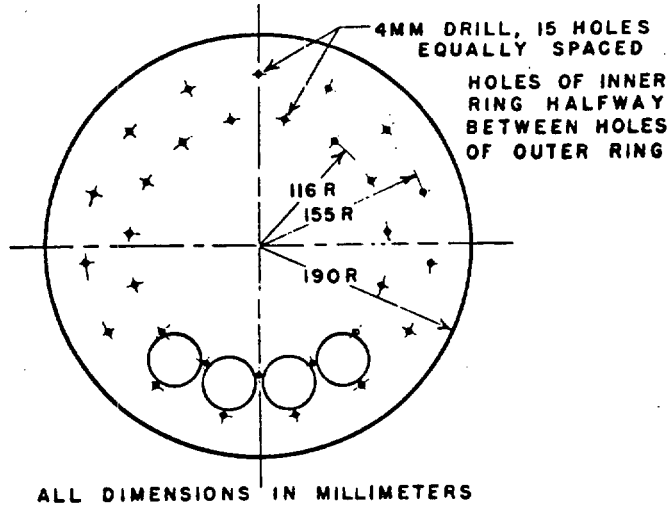


FIGURE 4

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LAMP TO TEST FOR COLOR STABILITY



ALL DIMENSIONS IN MILLIMETERS

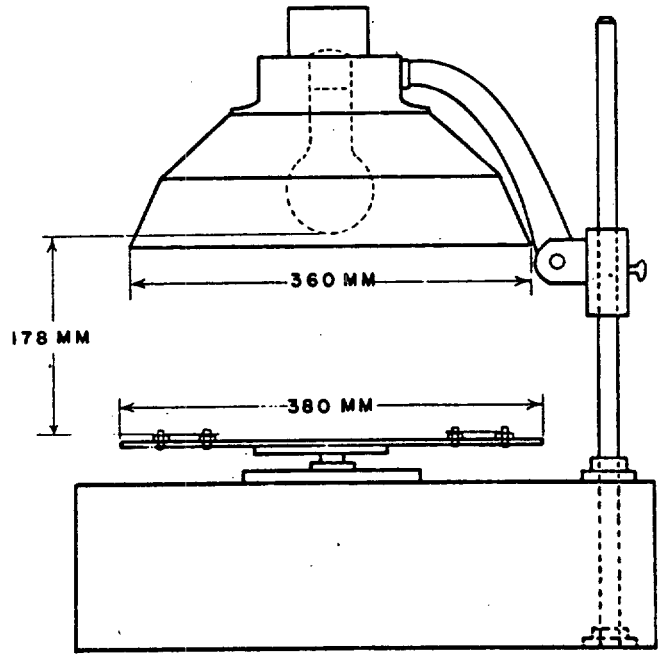


Figure 5

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

Fold

DEFENSE PERSONNEL SUPPORT CENTER
2800 SOUTH 20TH STREET
PHILADELPHIA, PA. 19101

DEFENSE SUPPLY AGENCY
OFFICIAL BUSINESS

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL
2800 SOUTH 20TH STREET
PHILADELPHIA, PA. 19101

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DPSC FORM 2418
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

U-R-00179a(DSA-DM)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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